

Disaster Response Planner: AI-Based Smart Emergency Response System with Intelligent Routing and Dispatch

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Abstract

Urban emergency response systems face significant challenges due to traffic congestion, delayed decision-making, and inefficient resource allocation. Traditional emergency dispatch systems rely heavily on manual coordination and static routing methods, which often fail to adapt to dynamic real-world conditions. This project presents an AI Smart Emergency Response System that integrates machine learning, deep learning concepts, and graph-based routing to automate and optimize the emergency response process.

The system accepts user input in the form of location and emergency type (medical, fire, or police) through an interactive Streamlit interface. A traffic estimation module predicts congestion levels using a lightweight neural network-based approach, while a severity prediction model evaluates the urgency of the incident based on traffic conditions. A rule-based intelligent dispatcher then assigns the most appropriate emergency vehicle,

such as a standard ambulance, ICU unit, fire truck, or police vehicle.

For route optimization, the system utilizes OpenStreetMap data through OSMnx and applies NetworkX shortest-path algorithms to compute the most efficient path between the incident location and the destination facility. The entire process is visualized using an interactive Folium map, where real-time vehicle movement is simulated along the computed route. Additionally, key metrics such as traffic level, severity score, assigned vehicle, and estimated time of arrival are displayed in a user-friendly dashboard.

Experimental results demonstrate that the system successfully integrates multiple AI components to provide fast, automated, and visually interpretable emergency response decisions. The proposed solution highlights the potential of combining artificial intelligence with real-world mapping technologies to develop scalable and cost-effective smart city emergency management systems.

I. INTRODUCTION

Cities today are expanding at a rapid pace, and with this growth comes an increasing demand for efficient emergency services. Whether it is a medical crisis, fire incident, or law enforcement situation, the speed at which help reaches the location can directly affect outcomes. However, in many urban areas, emergency response systems still depend on manual coordination, limited situational awareness, and static routing methods. These limitations often lead to delays, especially in traffic-heavy environments where quick decision-making is critical.

One of the major challenges faced by emergency services is navigating through unpredictable traffic conditions. Congestion, road blockages, and inefficient route selection can significantly slow down response time. Traditional dispatch systems usually rely on predefined routes or human judgment, which may not always reflect the current state of the road network. As a result, there is a growing need for intelligent systems that can adapt dynamically to real-time conditions and assist in faster decision-making.

Advancements in artificial intelligence have opened new possibilities for improving emergency response workflows. Machine learning techniques can be used to estimate the urgency of a situation based on available data, while lightweight predictive models can provide insights into traffic conditions. At the same time, graph-based algorithms applied to real-world map data make it possible to compute optimal routes quickly and accurately. When combined, these technologies can form a system that not only reduces response time but also improves the overall efficiency of emergency services.

Another important aspect of emergency management is selecting the right type of response unit. Not all incidents require the same level of resources, and sending either insufficient or excessive support can create operational inefficiencies. Intelligent decision systems can assist in assigning appropriate vehicles—such as ambulances, fire trucks, or police units—based on the nature and urgency of the situation. This ensures better utilization of available resources while maintaining responsiveness.

In this project, an AI Smart Emergency Response System is developed to address these challenges through a unified approach. The system accepts user input in the form of location and emergency type, processes this information through multiple AI-driven modules, and generates key outputs including traffic level, severity estimation, vehicle assignment, and optimal route. The integration of mapping tools allows the system to visualize the route and simulate the movement of the emergency vehicle in real time, providing a clear and interactive understanding of the response process.

The goal of this work is not to replace existing emergency infrastructure but to demonstrate how intelligent technologies can enhance it. By combining machine learning, routing algorithms, and interactive visualization within a simple web-based interface, the proposed system offers a practical and scalable foundation for future smart city emergency management solutions.

II. LITERATURE SURVEY

The application of artificial intelligence in emergency response and intelligent transportation has been widely explored in recent years. Researchers have focused on improving decision-making speed, optimizing routing, and enhancing situational awareness using data-driven approaches. These efforts highlight how AI can support critical operations where time and accuracy are essential.

A significant area of research involves traffic-aware routing for emergency services. Several studies have shown that incorporating traffic conditions into route planning can greatly reduce travel time compared to static navigation systems. Instead of relying on fixed routes, modern approaches use graph-based algorithms on real-world road networks to compute optimal paths dynamically. Techniques based on shortest-path algorithms, such as Dijkstra's method, have proven effective when combined with map data from platforms like OpenStreetMap. These approaches form the backbone of many intelligent navigation systems used today.

Another important research direction is the prediction of emergency severity using machine learning. Traditional systems often depend on manual judgment to assess the urgency of a situation, which can vary between operators. To address this, researchers have developed models that use historical data and contextual features to estimate the seriousness of incidents. Methods such as decision trees, random forests, and regression-based models have been used to classify emergencies into different priority levels. These models help in making faster and more consistent decisions, especially in high-pressure situations.

In addition to prediction, resource allocation has also been studied extensively. Assigning the correct emergency vehicle based on the situation is crucial for effective response. Some works have explored rule-based systems, while others have experimented with intelligent agents and optimization techniques to manage multiple resources. These systems aim to ensure that critical cases receive high-priority units while maintaining overall efficiency in resource utilization.

Recent advancements have also emphasized the integration of multiple technologies into a single system. Instead of treating routing, prediction, and dispatch as separate tasks, modern solutions attempt to combine them into a unified framework. Visualization tools and dashboards are often included to present real-time information in an easy-to-understand format. Interactive interfaces not only assist operators but also improve transparency and usability of the system.

Despite these developments, many existing solutions focus on individual components rather than a complete end-to-end system. Some concentrate only on route optimization, while others address severity prediction or vehicle allocation independently. The key limitation is the lack of integration between these modules.

The system proposed in this project addresses this gap by bringing together traffic estimation, severity prediction, intelligent dispatch, and real-time route visualization into a single platform. By combining these elements, the project demonstrates a more holistic approach to emergency response,

making it suitable as a foundational model for future smart city applications.

III. PROPOSED SYSTEM

The proposed AI Smart Emergency Response System is designed to improve the speed and efficiency of emergency handling by combining multiple intelligent components into a single workflow. Instead of relying on manual decision-making or fixed procedures, the system automates key steps such as traffic estimation, severity analysis, vehicle selection, and route planning. The goal is to provide a faster and more reliable response in critical situations.

At a high level, the system begins with user input, where a person enters their location and selects the type of emergency (medical, fire, or police) through a simple web interface. This input acts as the starting point for the entire process. The location is converted into geographic coordinates using a geocoding service, which allows the system to work with real-world maps.

Once the location is obtained, the system performs traffic estimation. A lightweight predictive model is used to generate a traffic level, representing how congested the roads are near the incident. This step helps simulate real-world conditions where traffic plays a major role in response time.

The next step is severity prediction. Based on the estimated traffic conditions, a machine learning model determines how critical the situation is. For example, higher traffic levels may indicate a more urgent situation due to possible delays. This severity score helps in prioritizing the response.

After that, the system moves to the dispatch stage, where a rule-based intelligent module selects the most appropriate emergency vehicle. The decision depends on both the emergency type and the severity level. For instance, a high-severity medical case may result in dispatching an ICU ambulance, while lower severity cases may use a standard ambulance. Fire and police emergencies are handled similarly with their respective vehicles.

The system then performs route optimization using real-world road network data. It uses mapping libraries to download the actual street network and applies shortest-path algorithms to find the most efficient route between the incident location and the destination (such as a hospital). This ensures that the path followed is realistic and suitable for navigation.

To make the system more interactive and understandable, a visualization module is included. An interactive map displays the calculated route, along with markers for the emergency location, destination, and the moving emergency vehicle. The movement is simulated step-by-step, giving a real-time feel of how the vehicle progresses toward the destination. Additional details such as traffic level, severity score, selected vehicle, and estimated time of arrival are also shown on the dashboard.

The system is built using a modular design, where each component—traffic prediction, severity analysis, dispatch logic, routing, and visualization—is developed separately but connected through a central application. This structure makes it easier to update or improve individual parts without affecting the entire system.

By combining all these functionalities into one platform, the proposed system provides a complete and practical solution for emergency response simulation. It demonstrates how artificial intelligence and mapping technologies can work together to support faster decision-making and better resource management in smart city environments.

IV. METHODOLOGY

It follows AI Smart Emergency Response System follows a structured sequence of steps to simulate how an intelligent emergency system operates in real-world conditions. Instead of relying on a single model, the system combines multiple components that work together—from receiving user input to generating route visualization. The overall workflow includes location processing, traffic estimation, severity prediction, vehicle assignment, route computation, and result visualization.

A. Location Processing and Input Handling

The process begins when the user enters a location and selects the type of emergency through the interface. Since the system needs geographical data to perform routing, the entered location is converted into latitude and longitude coordinates using a geocoding service. This step ensures that the system can accurately map the incident to a real-world position. If the entered location cannot be found, a default location is used to maintain system continuity.

B. Traffic Estimation

After obtaining the location, the system estimates the traffic conditions around the area. A lightweight predictive model is used to simulate traffic levels, producing a numerical value that represents congestion intensity. This step mimics real-world scenarios where traffic conditions directly affect emergency response time. The estimated traffic value is then passed to the next stage for further analysis.

C. Severity Prediction

The severity of the emergency is determined using a machine learning-based approach. The system takes the traffic level as input and predicts how critical the situation might be. Higher traffic values generally correspond to higher severity due to increased delays and potential risk. The output is a severity score that helps in prioritizing the response. This automated prediction reduces dependency on manual judgment and ensures consistency in decision-making.

D. Vehicle Assignment (Dispatch Logic)

Once the severity is calculated, the system selects an appropriate emergency vehicle using a rule-based decision module. The logic considers both the emergency type and the severity score. For example:

Medical emergencies → Ambulance or ICU unit

Fire emergencies → Fire truck

Police cases → Police vehicle

In high-severity situations, more advanced units such as ICU ambulances are assigned. This step ensures that the resources used are

suitable for the situation without unnecessary allocation.

E. Route Optimization

To determine how the emergency vehicle should travel, the system computes the shortest path between the incident location and the destination. It uses real-world map data obtained from OpenStreetMap and applies graph-based algorithms to find the optimal route. This approach ensures that the path follows actual roads instead of straight-line approximations. The output is a sequence of coordinates that represent the best possible route.

F. Estimated Time of Arrival (ETA)

The system also calculates an approximate arrival time based on traffic conditions. Instead of using complex real-time data, it applies a simple mapping between traffic level and travel time. Lower traffic results in shorter estimated time, while higher congestion increases the ETA. This provides users with a quick understanding of how long the response may take.

G. Visualization and Simulation

Finally, the system presents all results through an interactive dashboard. Key outputs such as traffic level, severity score, selected vehicle, and ETA are displayed clearly. In addition, a map visualization shows the computed route along with markers for the emergency location and destination.

To enhance understanding, the system simulates the movement of the emergency vehicle along the route. The vehicle's position updates step-by-step, giving a real-time effect of dispatch and travel. This visualization helps users and operators easily interpret the system's decisions.

V. CALCULATIONS AND METRICS USED IN THE SYSTEM

It follows AI Smart Emergency Response System, different calculations are performed to simulate real-world emergency handling. These calculations help in estimating traffic conditions, determining severity, selecting vehicles, and estimating travel time. Instead of relying on a single formula, the system uses a combination of simple mathematical relationships and logical mappings.

A. Traffic Level Estimation

Traffic is represented as a numerical value (usually between 0 and 100), indicating how congested the roads are. Since the current system uses a simulated or lightweight model, the traffic value is generated based on predefined logic.

A simple representation can be:

$\text{Traffic} \in [0, 100]$

Where:

Lower values $\rightarrow$ Free roads

Higher values $\rightarrow$ Heavy congestion

This value acts as the base input for further calculations.

B. Severity Calculation

The severity of the emergency is derived from the traffic level. The idea is that higher traffic may increase the urgency due to possible delays.

A simplified formula used in the system can be expressed as:

$\text{Severity} = \alpha \times \text{Traffic} + \beta$

Where:

$\alpha \rightarrow$ scaling factor

$\beta \rightarrow$ adjustment constant

In implementation, this is handled through a trained model or mapping function that converts traffic into a severity score between 0 and 100.

C. Vehicle Assignment Logic

Vehicle selection is not purely mathematical but follows conditional rules based on severity and emergency type.

This can be expressed as:

Vehicle =

{ICU Ambulance ,if Severity > 85 and Traffic > 70

Ambulance, if emergency = medical

Fire Truck, if emergency = fire

Police Vehicle, otherwise}

This ensures that critical situations receive advanced support.

D. Route Distance Calculation

The system calculates the shortest path between source and destination using graph-based routing.

Mathematically, this is:

Shortest Path = $\min \sum_{i=1}^n W_i \rightarrow$ distance (or weight) of each road segment

The total sum represents the minimum travel distance

This is computed using Dijkstra's algorithm on real road network data.

E. Estimated Time of Arrival (ETA)

ETA is calculated based on traffic conditions. Instead of exact physics-based modeling, a simple mapping is used.

$ETA = \text{Distance} / \text{Speed}$

Since speed depends on traffic:

$\text{Speed} = S_{\max} - k \times \text{Traffic}$

So,

$ETA \propto \text{Traffic}$

Meaning:

Higher traffic $\rightarrow$ Lower speed $\rightarrow$ Higher ETA

Lower traffic $\rightarrow$ Higher speed $\rightarrow$ Lower ETA

F. Progress Simulation

To simulate vehicle movement, the route is divided into multiple coordinate points.

$\text{Progress} = (\text{Current Step} / \text{Total Steps}) \times 100$

This value is used to update the progress bar and animate movement on the map.

VI. DECISION LOGIC AND SYSTEM EVALUATION (SIMPLIFIED CALCULATIONS)

In the proposed AI Smart Emergency Response System, decision-making is not based on a single calculation but on a combination of logical steps and computed values. These steps help the system decide how to respond to an emergency in a realistic and efficient way. Instead of fixed rules, the system adapts its behavior based on traffic conditions, severity level, and emergency type.

A. Decision Logic for Emergency Handling

After calculating traffic and severity, the system makes decisions in a structured way. The severity value is grouped into categories such as low, medium, and high. Based on this grouping:

Low severity $\rightarrow$ normal response

Medium severity $\rightarrow$ prioritized response

High severity $\rightarrow$ critical response

The system uses these categories along with the type of emergency to decide which vehicle should be sent. For example, a high-severity medical case results in dispatching a more advanced ambulance, while a lower severity case uses a standard one. This ensures that resources are used efficiently without overloading the system.

B. Emergency Priority Handling

One of the important features of the system is prioritization. In real-world situations, emergency cases must always be handled faster than normal conditions. In this project:

Emergency type automatically gets higher priority

High severity further increases urgency

Combined conditions lead to faster decisions and better resource allocation

This prioritization ensures that critical situations are not delayed due to general system logic.

C. Time Estimation Logic

The system estimates how long it will take for the emergency vehicle to reach the destination.

This is done using two main factors:

Distance of the route

Traffic level

Instead of using exact real-time speed data, the system adjusts travel time based on traffic conditions:

Low traffic → faster arrival

Medium traffic → moderate delay

High traffic → longer delay

This gives a realistic approximation of response time.

D. Route Progress Calculation

To simulate real-time movement, the route is divided into multiple small steps. The system tracks how far the vehicle has traveled by comparing:

Current position in the route

Total number of route points

This helps in:

Updating the progress bar

Animating the vehicle movement on the map

As the vehicle moves forward step-by-step, the system continuously updates the display to show progress.

E. System Performance Evaluation

Instead of traditional model evaluation (like accuracy or precision), this project focuses on functional performance, meaning how well the system works as a whole.

The evaluation is based on:

Correct handling of user input

Successful conversion of location to coordinates

Proper traffic and severity generation

Accurate vehicle assignment

Valid route generation using maps

Smooth visualization and simulation

During testing, the system was able to:

Handle different emergency types correctly

Generate consistent outputs for traffic and severity

Assign appropriate vehicles in all cases

Display routes and simulate movement without errors

F. Overall System Efficiency

The system is designed to be lightweight and fast. Since it uses simple models and logical mappings:

Processing time is low

Results are generated quickly

Interface responds smoothly

This makes it suitable for simulation and demonstration purposes.

IX. IMPLEMENTATION DETAILS

The AI Smart Emergency Response System is implemented using Python and a combination of libraries that support machine learning, mapping, and interactive visualization. The system is designed in a modular way so that each part—prediction, routing, and interface—works independently but connects smoothly to form a complete application.

At the core of the system is a Streamlit-based web application, which acts as the user interface. It allows users to enter their location and select the type of emergency. The interface is simple and interactive, making it easy to trigger the emergency response process and view results such as traffic level, severity, assigned vehicle, and estimated time of arrival.

For handling location input, the system uses a geocoding service to convert place names into latitude and longitude coordinates. This step is essential because all routing and map-based operations require geographic coordinates rather than text input.

The traffic estimation and severity prediction modules are implemented using lightweight machine learning logic. The traffic value is generated through a predictive function, and this value is passed to a severity prediction model built using libraries like scikit-learn. The output is a severity score that represents how urgent the situation is.

The dispatch module is implemented using a rule-based approach. Based on the emergency type and severity level, the system selects the most appropriate vehicle. This logic is written using simple conditional statements, ensuring transparency and ease of modification.

For route computation, the system uses OSMnx and NetworkX. OSMnx is responsible for downloading real-world road network data, while NetworkX applies graph-based algorithms to find the shortest path between the emergency location and the destination. This ensures that the generated route follows actual roads.

The map visualization is handled using Folium. It creates an interactive map where:

The emergency location is marked
The destination (such as a hospital) is shown
The computed route is drawn
The emergency vehicle is animated moving along the route

To simulate real-time movement, the route is divided into multiple points, and the vehicle marker is updated step-by-step.

The system also includes helper functions for tasks like estimating arrival time and converting route data into usable coordinates. These functions simplify the main application logic and improve code organization.

Overall, the implementation is structured into separate modules such as:

- Prediction (traffic and severity)
- Dispatch (vehicle assignment)
- Routing (path calculation)
- Visualization (map and dashboard)

This modular structure makes the system easy to understand, maintain, and extend in the future.

Component	Technology Used
Programming Language	Python

User Interface	Streamlit
Machine Learning Model	Scikit-learn
Deep Learning Framework	TensorFlow (basic usage)
Mapping & Routing	OSMnx, NetworkX
Map Visualization	Folium
Geocoding Service	Geopy (Nominatim)
Data Handling	Pandas
Numerical Computation	NumPy

X. RESULTS AND DISCUSSION

The proposed AI Smart Emergency Response System was tested using multiple input scenarios to evaluate its performance in handling emergency requests. The system was able to successfully process user-provided locations and different types of emergencies, demonstrating consistent and reliable behavior across all modules.

The location input provided by the user was accurately converted into geographic coordinates using the geocoding service. This allowed the system to work with real-world map data for further processing. The traffic estimation module generated values within a realistic range, and the severity prediction responded appropriately based on these traffic conditions. This shows that both modules are logically connected and produce meaningful outputs.

The vehicle dispatch module performed correctly by assigning suitable emergency vehicles based on the emergency type and severity level. For example, high-severity medical cases resulted in ICU ambulance assignment, while fire and police emergencies triggered their respective vehicles. This confirms that the decision logic is functioning effectively.

The routing module successfully retrieved real-world road networks and computed the shortest path between the incident location and the destination. The generated routes followed actual road structures, making the simulation realistic. The system was able to handle different locations without errors, indicating robustness in route computation.

One of the key outcomes of the system is the map-based visualization and simulation. The interactive map clearly displayed:

- Emergency location
- Destination point
- Computed route
- Moving emergency vehicle

The vehicle movement was simulated step-by-step along the route, providing a real-time feel of the emergency response. The progress bar and continuous updates improved user understanding of how the vehicle reaches the destination.

The system also displayed important metrics such as traffic level, severity score, assigned vehicle, and estimated time of arrival. The ETA values adjusted according to traffic conditions, giving a realistic approximation of response time.

Overall, the system performed efficiently with smooth interaction and minimal delay. All modules—including prediction, dispatch, routing, and visualization—worked together without conflicts. The results demonstrate that the proposed system can effectively simulate an intelligent emergency response process and serve as a foundation for future real-world applications.



XI. ADVANTAGES

The proposed AI Smart Emergency Response System offers several benefits compared to traditional emergency handling methods. One of the main advantages is the automation of decision-making, where the system processes user input and generates outputs such as severity, vehicle assignment, and route without manual intervention. This reduces human effort and minimizes delays during critical situations.

Another key advantage is the integration of multiple intelligent components into a single platform. The system combines traffic estimation, severity prediction, dispatch logic, and route optimization, which are usually handled separately. This unified approach improves overall efficiency and ensures faster response.

The use of real-world map data for routing makes the system more practical and realistic. Instead of using approximate paths, it calculates routes based on actual road networks, which helps in better simulation of emergency travel.

The system also provides a clear and interactive visualization through a dashboard. Important details such as traffic level, severity score, assigned vehicle, and estimated time of arrival are displayed in an easy-to-understand format. The map-based simulation of vehicle movement further enhances user understanding and engagement.

Another advantage is the modular design of the system. Each component—prediction, dispatch, routing, and visualization—is

implemented separately, making it easier to update or improve individual parts without affecting the entire system. Additionally, the system is lightweight and cost-effective, as it uses open-source tools and does not depend on expensive infrastructure. This makes it suitable for academic projects and scalable for future development.

Overall, the system provides a simple yet effective approach to demonstrate how artificial intelligence can improve emergency response in smart city environments.

XII. LIMITATIONS

Despite its advantages, the proposed system has certain limitations that should be considered. One of the main limitations is the use of simulated or simplified data for traffic and severity prediction. Since the system does not use real-time traffic APIs or real emergency datasets, the results are approximate rather than fully accurate.

Another limitation is that the system currently handles only one emergency request at a time. In real-world scenarios, multiple emergencies can occur simultaneously, requiring more advanced resource management and prioritization strategies.

The system also uses a fixed destination location (such as a predefined hospital). In practice, emergency services would need to select from multiple nearby facilities based on availability, distance, and specialization.

Dependence on internet connectivity is another constraint. Features such as geocoding, map loading, and road network retrieval require an active internet connection, which may not always be reliable during emergencies.

Additionally, the ETA calculation is simplified and does not consider real-time speed, road conditions, or dynamic changes like accidents or roadblocks. This limits the accuracy of arrival time predictions.

Finally, while the system provides a good simulation, it is not yet designed for real-world deployment, as it lacks integration with live traffic systems, GPS tracking, and emergency service databases.

XIII. FUTURE WORK

The proposed AI Smart Emergency Response System can be further improved by extending its capabilities beyond a single emergency scenario. One major enhancement would be supporting multiple simultaneous emergency requests. This would require intelligent resource management to prioritize cases and allocate available vehicles efficiently across different locations.

Another important improvement is the integration of real-time traffic data using external APIs. Instead of relying on simulated traffic values, the system can fetch live congestion data, which would make severity prediction, route planning, and ETA estimation more accurate and practical.

The system can also be extended to include multiple destination facilities such as hospitals, fire stations, and police stations. Instead of using a fixed location, the system can automatically select the nearest or most suitable facility based on distance, availability, and type of emergency.

In terms of intelligence, more advanced machine learning models can be incorporated. For example, severity prediction can be enhanced by considering additional factors such as time of day, weather conditions, and historical incident data. Similarly, optimization techniques or AI-based decision models can be used to improve vehicle dispatch in complex scenarios.

Another possible extension is the development of a mobile application or real-time tracking system. This would allow users and authorities to monitor emergency vehicles using GPS and receive live updates during the response process.

Finally, the system can be integrated with IoT and smart city infrastructure, such as traffic signals and road sensors. This would enable features like automatic traffic signal clearance for emergency vehicles and better coordination across the city.

XIV. CONCLUSION

This project presented an AI Smart Emergency Response System that combines machine learning, routing algorithms, and interactive visualization to simulate an intelligent emergency handling process. The system automates key steps such as traffic estimation, severity prediction, vehicle assignment, and route calculation, reducing the need for manual intervention.

Unlike traditional approaches, the system provides a complete end-to-end solution, where all components work together in a single platform. It uses real-world map data to compute accurate routes and displays results through a user-friendly dashboard. The inclusion of map-based simulation further improves understanding by showing how the emergency vehicle moves toward the destination.

The results demonstrate that the system performs reliably across different scenarios, producing consistent outputs for traffic, severity, and vehicle selection. The modular design makes it flexible and easy to extend, allowing future improvements without major changes to the existing structure.

Although the current system uses simplified models and simulated data, it successfully demonstrates the potential of applying artificial intelligence to emergency response systems. With further enhancements such as real-time data integration and multi-incident handling, the system can evolve into a practical solution for smart city applications.

Overall, this project provides a strong foundation for building intelligent, efficient, and scalable emergency response systems using modern technologies.

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